

Alexander Kurakin

Game Systems Engineer | Principal Software Engineer

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PROFESSIONAL SUMMARY

Principal Software Engineer with over a decade of professional experience in high-reliability systems, translating enterprise engineering into professional game development. Expert in C# architecture, Unity middleware integration, and scalable gameplay systems.

WORK EXPERIENCE

Oracle (NetSuite) | Principal Cloud Database Engineer

Apr 2020 – Present

Leveraging high-level engineering expertise to drive reliability, performance, and architectural standards across distributed systems.

- **Performance optimization:** expertly diagnosed and resolved bottlenecks in high-load systems using advanced tracing. Applied rigorous optimization strategies to legacy codebases, achieving performance parity across platforms.
- **Tooling and automation:** led the development of internal diagnostic tools and automated replay frameworks to stabilize legacy pipelines, significantly reducing debugging time for the wider engineering team.
- **Engineering leadership:** mentored senior engineers and directed cross-functional initiatives. Championed strict version control (Perforce/Git) and defensive programming standards.

Integro Technologies / Seertech Solutions | Lead Technical Expert & Developer

Apr 2016 – Apr 2020

Full-stack development involving system design, legacy modernization, and cross-stack integration.

- **Systems architecture:** led the modernization of critical legacy modules for major enterprise clients, enforcing separation of concerns and improving system stability.
- **Cross-language integration:** engineered complex data pipelines involving JSON processing and RESTful APIs. Debugged C# and Java interoperability issues to ensure seamless data flow.
- **Technical management:** promoted to Team Lead within 6 months to oversee project delivery, conduct technical interviews, and define architectural standards.

Independent Developer | Game Systems Engineer | Full Stack Developer

Jan 2016 – Present

Continuous development of commercial mobile titles, technical demos, and complex gameplay systems.

- Shipped **Parking Master: Low-Poly** to Google Play Store, managing full lifecycle from development to iterative updates.
- Engineered **5yearsApp2D**, a cross-platform Unity application with a .NET/NoSQL backend (Firebase)
- Developed custom tools for audio professionals (**CamelotOverlay**) and middleware integration workflows.
- Engineered separate R&D projects for NavMesh AI, inventory system, and event-driven logic.

GAME DEVELOPMENT PROJECTS

Digital Hoarder | Jan 2024 – present

Flagship systemic 3D action-adventure game focusing on exploration and non-violent problem solving.

- Architected a modular codebase allowing seamless addition of mechanics without regression.
- Engineered robust third-person traversal, puzzle systems, and world-state driven quest engines.
- Developed a dedicated 'Test room' environment for regression testing.

Gameplay Systems R&D | Oct 2025 – Apr 2026

A suite of decoupled systems focusing on clean code, architecture, and reusability.

- Designed communication layers using C# events/delegates to remove dependencies.
- Created tactical AI agents utilizing NavMesh traversal and physics-driven movement.
- Independently designed and implemented gameplay loops based on various conditions.

Tooling and middleware | Jan 2019 – Dec 2024

Custom C# visual tools for audio professionals and middleware integration.

- Designed a visual utility for audio professionals visualizing harmonic compatibility (extended Circle of Fifths) to assist DJs with musically consonant transitions.
 - Orchestrated interoperability of complex middleware (GKC, Malbers, RCCP) and authored documentation.
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TECHNICAL SKILLS

- **Languages:** C# (Expert), Python, SQL, PL/SQL, Delphi/Object Pascal, Java.
 - **Unity Engine:** Physics, Vector math, AI navigation, Animation (Humanoid/Blend Trees/IK), UI prototyping, Cinemachine/Timeline, SFX and VFX setup.
 - **Architecture:** Event-driven architecture, OOP/MVC, Asynchronous programming, Facade / Strategy / Open/close patterns.
 - **Tools and DevOps:** Git / SVN / Perforce / Teamcity / Nexus, Unity Profiler / ADB logcat, Visual Studio, ElasticSearch / OpenSearch / Prometheus / Graphana
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EDUCATION

Master's Degree | Information Systems and Technologies

Moscow Aviation Institute (National Research University)

2007 – 2015

- Structured programming (Delphi/Object Pascal), Database architecture, Algorithms & data structures, Systems engineering.
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PUBLICATIONS

Game Kit Controller (GKC) Integration documentation | Apr 2025 | [link](#)

Collaborated with the GKC lead developer to diagnose critical configuration steps and authored the official integration workflow update for the wider community.

INTERESTS

Game Design: Systemic gameplay, Emergent narratives, Clean code, Tool development

Music: Proud owner of 5 different musical instruments, Proficient Ableton Live user, Playing by ear and experimenting for fun

Videogames: Lifelong passion from 8bit and IBM era, RPG as the favourite genre, Backlog of ~150 titles